# GanJiang User Manual

Powder X ray diffraction analysis and crystal structure operations

## Scope

This manual covers task creation, single and multiple phase identification, refinement, PDF card calculation, XRD simulation and crystal viewing. Each procedure explains the entry point, required inputs, confirmation, execution and saving.

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# 01 Function navigation

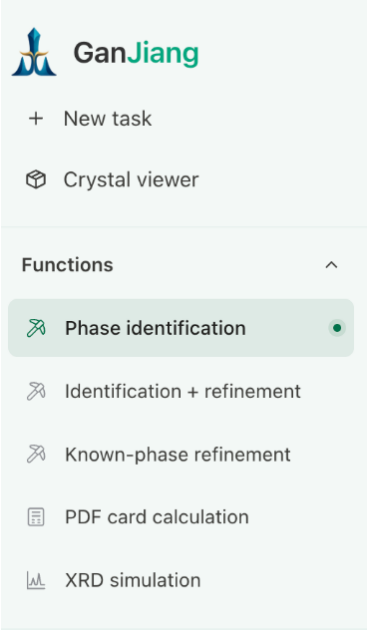



**Figure 1 Function navigation in the English interface.**

Select the required function on the left. Use identification or refinement to analyse an observed pattern. Use XRD simulation to calculate diffraction from a structure. Crystal viewer displays the atomic arrangement in a CIF.

PDF card calculation compares theoretical reflections from supplied structures with observed peaks. Here PDF means powder diffraction card, not a PDF document or pair distribution function analysis.

# 02 Files and measurement conditions

## Prepare attachments by function

| Function | Required input |
| --- | --- |
| Phase identification | One observed pattern; optional elements or formula hints |
| Identification + refinement | One observed pattern; subsequent refinement confirmation |
| Known-phase refinement | One observed pattern and a CIF for each phase |
| PDF card calculation | One observed pattern and at least one CIF |
| XRD simulation | One or more CIF/MCIF files; no observed pattern required |
| Crystal viewer | One CIF file |

## Pattern format

The identification workflow accepts CSV, DAT, RAW, TXT, XRDML and XY. Submit one observed pattern per task. If an instrument file cannot be parsed, export a two-column CSV or XY file from the instrument software; changing its extension does not convert it.

A two-column text pattern normally contains 2θ in degrees and intensity, with one point per row. Preserve the row pairing and state the header and any background subtraction, smoothing, normalization or $K\alpha_2$ removal.

## Measurement conditions

Record the sample ID, radiation source or wavelength, axis units, scan range and available instrument file. The current identification/refinement test interface is fixed to Cu Kα. Do not relabel non-Cu data as Cu data, and do not treat a software default as a measured condition.

# 03 Create a task and confirm settings

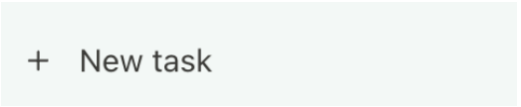



**Figure 2 Use New task for an independent sample.**

## Submission procedure

1. Click New task and select the function on the left.

2. Use “+” in the bottom task input area to attach the individual files. Check their names; do not upload a whole test archive as one pattern.

3. Enter the objective, sample ID and known conditions in the same input area. Add to the prefilled request as needed. The chat model selector is not the identification engine or radiation setting.

4. Send the request using the lower-right button. When the parameter form appears, review the visible task, phase mode, retrieval models, elements and formula fields.

5. The submitted form values control the calculation. Submit the completed form; closing or cancelling it does not start computation.

6. Read the task execution messages. If the plan is awaiting confirmation, complete that form before uploading the same inputs again.

## Request text and confirmation forms

The request expresses intent; the form saves the final calculation settings. Even if the request says multiphase, verify the phase mode in the form. Naming two models does not establish that both services ran successfully; check the execution record.

Continue the existing task when supplying missing information for the same sample. Create a separate task for another sample or an independent setting comparison, and include the sample ID and conditions in its name.

# 04 Single phase identification

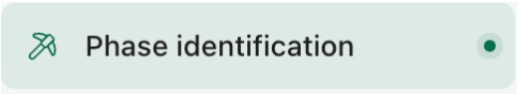



**Figure 3 Phase identification entry.**

## Purpose and input

Search for candidate crystalline phases from one observed pattern without refinement. A CIF is not required for unknown-phase identification. Leave formula hints empty when testing identification without formulas.

## Single phase settings

| Option | Role |
| --- | --- |
| Single phase | Search under a single-phase assumption; retain unexplained peaks. |
| XMatcher | Pattern-based candidate matching; can be used without element constraints. |
| XQueryer-LW | Candidate retrieval using valid user-supplied element constraints. |
| Both models | Select both in the single-phase form, then verify actual execution. |
| Unknown phase count | Try a single phase first; fall back to multiphase analysis if quality is insufficient. |

## Example request

Identify a single phase in the attached pattern without refinement.

Known elements: Ca Fe Si O

No formula hints are provided. Use XMatcher and XQueryer-LW in the single-phase configuration.

Radiation: [actual conditions]. Preprocessing: [actual processing or none].

## Completion checks

Check retrieval success before candidate ranking and validation. Top 1 means the first ranked candidate, not an automatically confirmed phase. If no candidate list was generated, there is no Top 1 to select.

# 05 Multiphase identification

## Multiphase configuration

Multiphase means multiple crystalline phases in one sample pattern, not multiple uploaded patterns. Select Phase identification, attach one mixed pattern and choose the multiphase mode in the confirmation form. Use unknown phase count when the count is not established.

The multiphase form offers XMatcher-AutoMix and XDecomposer. Keep both default channels or select one for a controlled test. Candidates from both are pooled for validation. The single-phase XQueryer-LW and XMatcher choices do not replace the multiphase settings.

## Enter elements separately from formulas

| Field | How to enter it |
| --- | --- |
| Elements | Enter unique known symbols, e.g. Ca Al Si O Ni S. Do not guess unknown elements; identify a partial list as incomplete. |
| Formula hints | Leave empty if unavailable. For known phases, enter one formula per phase, separated by ASCII commas, semicolons or newlines. |
| CIF files | For an element-constrained identification test without structure hints, upload the pattern only and retain CIFs as reference answers. Attach them for known-phase refinement. |

## Example request

Identify multiple phases in the attached pattern without refinement or formula hints.

Known elements: Ca Al Si O Ni S

Use XMatcher-AutoMix and XDecomposer in the multiphase configuration.

Radiation: [actual conditions]. Phase count: [unknown or established count].

Before submission, check: one pattern, multiphase mode, correct elements and an empty formula field. “Not reviewed” means validation is pending or missing; it does not mean the candidates have been rejected.

# 06 Identification and refinement

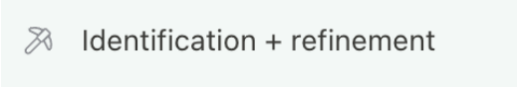



**Figure 4 Identification + refinement entry.**

## Purpose and workflow

Use this function to find phases and then test their structural models against the observed pattern. Provide one pattern and measurement conditions; known elements constrain identification. Identification and refinement are separate stages and must be assessed separately.

1. Select Identification + refinement, attach the pattern and specify single, multiple or unknown phase count.

2. Submit the identification form after checking elements, formula hints and channels.

3. Review the identification candidates and evidence. In the refinement form, check the selected phases or structural files.

4. Select an available engine and provide any instrument file, fit range, constraints and round settings. Do not release all atomic parameters simply to improve a fit.

5. Submit refinement confirmation and keep the task session available. After a transport interruption, check the existing job before submitting another calculation.

6. Review observed/calculated profiles, differences, fit metrics, structure files and validation status; download this task's outputs.

## Example request

Identify phases in the attached pattern and refine supported candidates.

Sample: [ID]. Elements: [known elements or unknown]. Radiation: [actual conditions].

Start with scale, background and cell-related parameters; fix atomic occupancies.

Provide the whole-pattern fit, difference profile, fit metrics and final validation status.

# 07 Known phase refinement

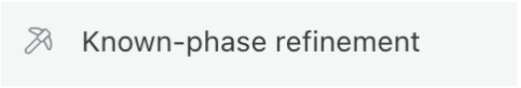



**Figure 5 Known-phase refinement entry.**

## Purpose and input

Use supplied structural models directly, skipping phase retrieval. Attach one CIF for a single phase, or a CIF for every phase in a mixture, together with one complete observed pattern.

1. Select Known-phase refinement and attach the pattern and all CIFs using “+”.

2. Check each CIF’s formula, cell and atom information. Avoid duplicate structures and missing phases.

3. Confirm phase mode, structure list, engine and instrument configuration in the refinement form. Use the corresponding IRF file when supplying a FullProf instrument profile.

4. Define the variables to fit. Begin with scale and background, then add cell and profile variables as justified by the data.

5. Submit and check that all phases entered the model. Preserve failure messages and do not label a partial result as fully successful.

| Parameter | Role and check |
| --- | --- |
| Scale factors | Control phase intensity; not themselves mass fractions. |
| Background and profile | Fit baseline and peak shape; do not absorb missing phases into the background. |
| Zero offset and cell | Affect peak positions; consider parameter correlations. |
| Coordinates and occupancies | Affect structure factors; vary only when supported by data and chemistry. |

For phase fractions, check the definition, included phases and normalization. Display scales, candidate scores and refined mass fractions are different quantities.

# 08 PDF card calculation

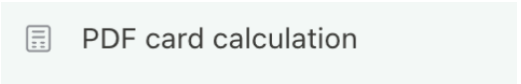



**Figure 6 PDF card calculation entry.**

Supply one observed pattern and one or more known CIF structures to calculate theoretical reflections and compare them with observed peaks. This function does not perform unknown-phase identification or Rietveld refinement.

1. Select PDF card calculation and attach the pattern and all phase CIFs.

2. Check filenames, wavelength and peak detection/matching parameters in the form.

3. Submit. Review the theoretical reflection table and observed peak table separately, including assignments and unmatched peaks.

4. When changing visibility, colours or annotations, retain phase names and sources. After editing assignments, export the current version using the report controls.

| Parameter | Initial value | Meaning |
|---|---|---|
| Wavelength | 1.5406 Å | Radiation used for theoretical positions; match the experiment. |
| Matching tolerance | 0.20° 2θ | Allowed observed-to-theoretical position difference. |
| Intensity threshold | 10% | Relative intensity threshold for observed peaks. |
| Minimum distance | 0.08° 2θ | Minimum separation between detected peaks. |
| Prominence | 0% | Required prominence above the surrounding baseline. |

Initial values are software defaults and must be checked for the dataset. Download theory.csv, peaks.csv, observed.csv, parameters.json or the result bundle. Initial files contain automatic assignments; later manual edits require a new export.

A missing match does not by itself prove a phase is absent. CIF-derived reflections do not establish a commercial database card number.

# 09 XRD simulation

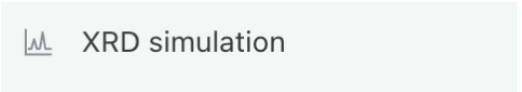



**Figure 7 XRD simulation entry.**

Calculate theoretical powder diffraction from CIF or MCIF files without an observed pattern. Upload one or more structures to compare component patterns and their total.

1. Select XRD simulation and attach the structures; separate files represent separate phases.
2. Check wavelength, 2θ range, sampling step, peak shape, width and mixing parameter in the form.
3. Submit and inspect parsing status for every file. If some files fail, only the successful phases are completed.
4. In the panel, adjust visibility, colour, relative display scale and width. Choose total, overlay or stacked views and inspect the HKL peak table.
5. Use the panel CSV or SVG export for the current view. A wavelength, range or sampling-step change requires a newly confirmed calculation.

| Setting | Initial value | Purpose |
|---|---|---|
| Wavelength | 1.5406 Å | Cu $K\alpha_1$ theoretical calculation |
| 2θ range | 10–90° | Simulation interval |
| Step | 0.02° | Angular interval for sampled curves |
| Shape | pseudo-Voigt | Peak profile function |
| FWHM | 0.18° | Initial peak broadening |
| Lorentzian fraction η | 0.5 | Profile mixing parameter |

Initial downloads include peaks.csv, pattern.csv, parameters.json and a ZIP, using the confirmed settings. Relative display scale is not mass fraction; simulation does not replace experimental identification or quantitative refinement.

# 10 Crystal viewer

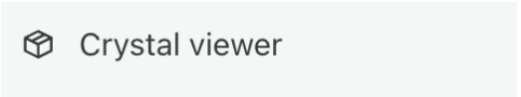



**Figure 8 Crystal viewer entry.**

## Open a structure

1. Click Crystal viewer to open the structure page.
2. Choose one .cif using the CIF file control, or drop it into the upload area if shown.
3. Wait for parsing and check the active filename and structure information. Load the new CIF when changing samples so the previous model is not mistaken for the current one.
4. Use the available view controls to inspect the atomic arrangement. Return to XRD simulation when a theoretical diffraction pattern is required.

## Read elements from a CIF

Open the CIF in a text editor and search for _chemical_formula_structural, _chemical_formula_sum or _atom_site_type_symbol. Element symbols identify the species; numbers describe stoichiometry. An expanded cell formula may have larger numbers than a reduced structural formula while listing the same elements.

| Example formula | Elements |
|---|---|
| Ca3Al2(SiO4)3 | Ca Al Si O |
| NiS | Ni S |
| Union for the two phases | Ca Al Si O Ni S |

## If parsing fails

Confirm the file is extracted and is a genuine CIF, with complete cell and atom-site information. Renaming a CSV or image cannot create a structural file. A displayed model establishes that it was parsed, not that the phase was detected in the sample.

# 11 Status and task records

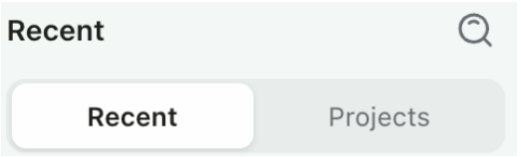



**Figure 9 Recent and Projects provide access to tasks.**

## Check execution status first

| Status or message | Meaning and next action |
|---|---|
| Awaiting confirmation | Computation has not started; submit the current parameter form. |
| Running or awaiting analysis | Not a final conclusion; inspect execution messages and progress. |
| Reviewed 0; not reviewed > 0 | Candidates exist without validation verdicts. If this persists after completion, preserve records and investigate validation. |
| XMatcher call failed | Retrieval failed technically; inspect the error. This is not evidence that the sample has no matching phase. |
| No reliable phase match found | No reliable conclusion is released. Examine channel status, validation and input conditions to determine why. |
| invalid formula_hint | Check formulas and separators; clear the field when no formulas are intended. |
| incomplete chunked read | The response ended during transfer. Check whether the original job is still running before recovery or retry. |

## Save and review

Return through Recent and verify the sample ID, attachments, settings and time. Retain inputs, confirmed settings, outputs and error records. Download the result attachments; screenshots capture visible content and do not replace full CSV tables or bundles.

A transport message alone cannot locate the fault in the network, model or compute service. For persistent failures, provide the task name, full error and execution record; repeated new tasks can obscure the original failure.